Trajectory Tracking Control for Quadrotor UAVs Based on a Finite-Time Extended State Observer

Yuxin Li, Yan Qi*

***Abstract*—This paper proposes a novel trajectory tracking control scheme for quadrotor UAVs. The main contributions and advantages of this work are summarized as follows: First, a new extended state observer is designed for quadrotor UAVs, which enables finite-time accurate estimation of disturbance signals. Second, command filtering control is adopted to avoid the differentiation of virtual control laws and effectively reduce the system's computational burden. Third, finite-time control theory is incorporated into the controller design to ensure the rapid response performance of the system. Finally, numerical simulation results verify that the proposed scheme achieves excellent control performance and exhibits strong anti-disturbance capability.**

***Index Terms*—quadrotor UAVs, extended state observer, command filter, finite-time control.**

I. Introduction

Unmanned Aerial Vehicles (UAVs) have become indispensable equipment in the fields of aerospace and military reconnaissance. In particular, quadrotor UAVs are the most widely adopted, owing to their outstanding agility and unique vertical takeoff and landing (VTOL) capabilities. It is worth noting that the quadrotor UAVs are underactuated, strongly coupled, and high-order systems, making the design of high-precision controllers a challenging task. Furthermore, the quadrotor UAVs work at high altitudes, addressing the effects of gust disturbances is a topic of practical significance [1]–[3].

Currently, there are many control schemes for the quadrotor UAVs, such as Proportional-Integral-Derivative (PID) control [4], fuzzy adaptive control [5], Sliding Mode Control(SMC) [6]–[8], and backstepping control [9]–[11]. PID control is used control scheme in early UAVs. In [4], a fuzzy PID technique is used to obtain the optimal control parameters in real time, thereby achieving trajectory tracking. As the demand for control precision continues to rise, relying solely on PID control can no longer meet the control requirements. SMC and backstepping control have gradually developed into commonly used control methods for quadrotor UAVs. In [6], the Neural networks are utilized to obtain the equivalent control and the switching control values in SMC, thereby reducing control costs. However, neural networks have errors when approximating nonlinear functions. To further reduce the chattering phenomenon, a super-twisting SMC was designed in [7] to suppress chattering in the input signals, and an adaptive law in [8] to compensate for external disturbance signals. Nevertheless, the chattering in SMC cannot be eliminated. In contrast, the control laws of backstepping control is a smooth function derived based on Lyapunov theory, which avoids the chattering problem. In [9], the backstepping control is applied to quadrotor UAVs which ensures the anti-noise capability of the system. In backstepping control, the differentiation of the virtual control law may lead to the phenomenon of differential explosion, which is usually addressed by the dynamic surface technique [10], [11]. However, dynamic surface technology relies on the characteristics of first-order filters to achieve simplified calculations. Its signal transitions are relatively gradual, making it difficult to apply efficiently in drones that demand high response speeds [12]. Therefore, based on the dynamic surface technique, it is of great necessity to further enhance the response speed of the control system.

It is also worth noting that quadrotor unmanned aerial vehicles often encounter external disturbances such as wind gusts and ground effects. If these disturbance signals cannot be compensated for in a timely manner, the UAV mission is highly likely to fail. Currently, constructing a disturbance observer based on feedforward control is recognized as one of the most effective solutions to this problem. [13], [14]. In [13], a sliding mode observer is proposed to ensure that disturbance signals are compensated within fixed-time. In [14], fuzzy logic is introduced into the disturbance observer, which simultaneously compensates for the lumped disturbances of the system. However, the approximation capability of fuzzy logic is limited, and the output of the sliding mode observer is not smooth, which will affect the control performance.

In addition, rapidity is a crucial indicator for evaluating system performance. Different from traditional infinite-time control, finite-time control ensures the system's rapidity by introducing fractional power terms to drive errors back to the equilibrium point. As demonstrated by the conclusions in [15]–[17], finite-time control can significantly reduce the system's convergence time, which provides important inspiration for the research in this paper.

Based on the above literature review, a new finite-time control scheme is proposed, the details of which are: A Finite-Time Extended State Observer(FTESO) is used to estimate disturbances, and the finite-time command filtering is introduced into the backstepping design process. The advantages of proposed scheme are:

(i) Compared with the observer in [13], the proposed observer does not exhibit chattering, has a simpler structure, and achieves the finite-time convergence of observation errors.

(ii) Finite-time theory is applied to the design of the observer, filter, and controller, ensuring that all variables can converge to the neighborhood of the equilibrium point in finite-time.

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Yuxin Li is an undergraduate student of Department of Electronic Information, University of Science and Technology Liaoning, Anshan, 114051, China (e-mail: 3694681873@qq.com).

Yan Qi is a lecturer of Department of Electronic Information, University of Science and Technology Liaoning, Anshan, 114051, China (corresponding author to provide phone: 15241831334, e-mail: 7623970@qq.com).

(iii) The hyperbolic tangent function is chosen as the saturation function to alleviate the actuator burden.

In Section II, the Euler angles modeling scheme for the quadrotor UAV is introduced. In Section III, the quadrotor UAV is decomposed into an inner-loop system and an outer-loop system. Control schemes for both the inner and outer loops are proposed separately. A global stability proof based on Lyapunov theory is proposed in Section IV. In section V, the effectiveness of the algorithm is verified through numerical simulations.

II. Model description

A. Model of quadrotor UAV

The structural diagram of a quadrotor UAV is shown in Fig.1. O-$b_1b_2b_3$ denotes a inertial coordinate system, which remains relatively stationary with respect to the ground. O-$c_1c_2c_3$ represents a body coordinate system that stays relatively stationary relative to the quadrotor UAV. The relative position of the two coordinate systems in three-dimensional space is denoted by x, y, z. ϕ is the roll angle, θ is the pitch angle, and ψ is the yaw angle. The angular velocities corresponding to the three Euler angles are denoted by p, q,and r respectively. The four rotors are driven by M_1, M_2, M_3, M_4, and the state variable is selected as $\mathbf{X} = [\dot{x}, \dot{y}, \dot{z}, x, y, z, p, q, r, \phi, \theta, \psi]^{\mathrm{T}} = [x_1, \cdots, x_{12}]^{\mathrm{T}}$.

Before introducing the mathematical model, the following assumptions need to be used.

Assumption 1 The structural parameters of the system are constant.

Assumption 2 To rule out extreme cases such as the UAV flipping, the roll angle ϕ and pitch angle θ of the UAV are limited to the range $(\frac{\pi}{2}, -\frac{\pi}{2})$.

Assumption 3 The external disturbance signals are continuous and differentiable.

The common modeling expressions of the quadrotor UAV are presented as

$$\begin{cases} \dot{x}_1 = \frac{[\cos(x_{10})\cos(x_{12})\sin(x_{11})+\sin(x_{10})\sin(x_{12})]\mathrm{s}(u_1)-k_t x_1}{m} \\ \dot{x}_2 = \frac{[(\cos_{x_{10}}\sin(x_{11})\sin(x_{12})-\cos(x_{12})\sin(x_{10})]\mathrm{s}(u_1)-k_t x_2}{m} \\ \dot{x}_3 = \frac{[\cos(x_{11})\cos(x_{10})]\mathrm{s}(u_1)-mg-k_t x_3}{m} \\ \dot{x}_4 = x_1 \\ \dot{x}_5 = x_2 \\ \dot{x}_6 = x_3 \\ \dot{x}_7 = \frac{-k_r x_7 - x_8 x_9 (I_x - I_y) + \mathrm{s}(u_2) + d_p}{I_x} \\ \dot{x}_8 = \frac{-k_r x_8 - x_7 x_9 (I_x - I_z) + \mathrm{s}(u_3) + d_q}{I_y} \\ \dot{x}_9 = \frac{-k_r x_9 (I_y - I_x) + \mathrm{s}(u_4) + d_r}{I_z} \\ \dot{x}_{10} = x_7 + x_8 \sin(x_{10})\tan(x_{11}) + x_9 \cos(x_{10})\tan(x_{11}) \\ \dot{x}_{11} = x_8 \cos(x_{10}) - x_9 \sin(x_{10}) \\ \dot{x}_{12} = \frac{x_8 \sin(x_{10}) + x_9 \cos(x_{10})}{\cos(x_{11})} \end{cases} \tag{1}$$

where u_1 is the lift force, u_2, u_3,and u_4 represent the input torques in three directions, k_r, k_t are the drag coefficient, m is the mass, and g is the gravitational acceleration. I_x, I_y, I_z are the moments of inertia, and they are all constants. d_p, d_q, and d_r, are used to describe disturbances such as gusts that may be encountered during flight and those caused by modeling dynamics. $\mathrm{s}(\cdot)$ is a saturation function of the system. In (1), the input signals are u_1, u_2, u_3,and u_4 and the output signals are $x, y, z, \phi, \theta, \psi$.

Remark 1 Most traditional schemes use the sign functions as the saturation functions [20], [21], which will lead to drastic changes in the system input. To ensure the system obtains a smooth input signal, a hyperbolic tangent-type saturation function is adopted in the proposed scheme. The specific form is

$$\mathrm{s}(x) = x_{\max} \tanh(\frac{x}{x_{\max}}) \tag{2}$$

where x represents an arbitrary variable, and its allowed maximum value is $x_{\max}$.

Our control objective is summarized as

(i) A disturbance observer is designed, which can provide the system with an estimated values approximating the actual disturbance within finite-time.

(ii) Ensure that both the flight trajectory and attitude angles of the quadrotor UAV can track desired signals, and all of the variables in the system can converge to the vicinity of the equilibrium point within finite-time.

In the process of controller design and proof, the following lemmas are required.

B. preparatory knowledge

Lemma 1 [18] For arbitrary $\mathbf{x} = [x_1, x_2, \cdots, x_n]^{\mathrm{T}}$, and $p \in (0, 1]$, the following inequality is ture

$$(\sum_{i=1}^{n} |x_i|)^p < \sum_{i=1}^{n} |x_i|^p < n^{1-p} (\sum_{i=1}^{n} |x_i|)^p \tag{3}$$

Lemma 2 [19] Consider the nonlinear system $\dot{x}(t) = f(x(t))$, if the Lyapunov function of the system satisfies:

$$\dot{V}(x) \leqslant -AV^s(x) + B \tag{4}$$

where $A > 0, B > 0, s \in (0, 1)$, the nonlinear system can converge to $\mathbf{X} = \{x | V^s \leqslant \frac{B}{A\nu(1-s)}\}$ in a finite-time. The convergence time T satisfies

$$T \leqslant \frac{V^{1-s}(t_0)}{A\nu(1-s)} \tag{5}$$

where $\nu \in (0, 1)$, t_0 denotes the initial time.

III. Controller design

It can be known from 1 that the system is an underactuated system. A double closed-loop control strategy is used to convert the system into a fully actuated system.

The inner loop system is selected as

$$\begin{cases} \dot{\mathbf{X}}_1 = \mathbf{A}_1 \mathbf{X}_2 \\ \dot{\mathbf{X}}_2 = \mathbf{A}_2 + \mathbf{B}_1 \mathrm{s}(\mathbf{u}_i) + \mathbf{D}_1 \end{cases} \tag{6}$$

where $\mathbf{X}_1 = [x_6, x_{10}, x_{11}, x_{12}]^{\mathrm{T}}$, $\mathbf{X}_2 = [x_3, x_7, x_8, x_9]^{\mathrm{T}}$, $\mathbf{u}_i = [u_1, u_2, u_3, u_4]^{\mathrm{T}}$, $\mathbf{B}_1 = diag(\cos(x_{10})\cos(x_{11})/m, 1/I_x, 1/I_y, 1/I_z))$, $\mathbf{D}_1 = [d_{11}, d_{12}, d_{13}, d_{14}]^{\mathrm{T}} = [0, \frac{d_p}{I_x}, \frac{d_q}{I_y}, \frac{d_r}{I_z}]^{\mathrm{T}}$.

$$\mathbf{A}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & \sin(x_{10})\tan(x_{11}) & \cos(x_{10})\tan(x_{11}) \\ 0 & 0 & \cos(x_{10}) & -\sin(x_{10}) \\ 0 & 0 & \frac{\sin(x_{10})}{\cos(x_{11})} & \frac{\cos(x_{10})}{\cos(x_{11})} \end{bmatrix} \tag{7}$$

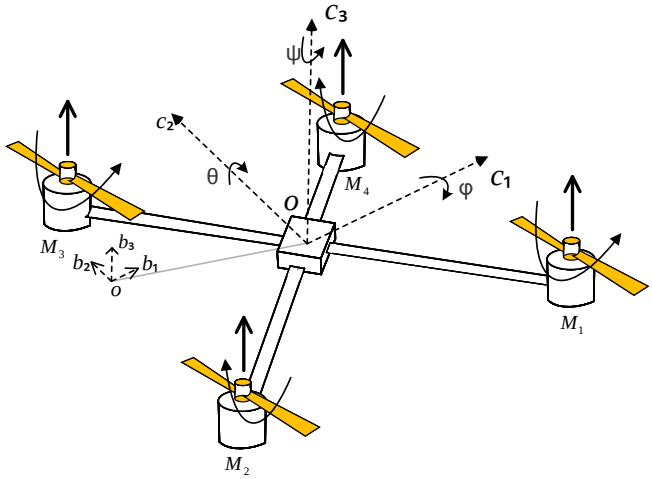

Fig. 1. Quadrotor UAV Model.

$$\mathbf{A}_2 = \begin{bmatrix} (-mg - k_t x_3)/m \\ [-k_r x_7 - x_8 x_9 (I_z - I_y)]/I_x \\ [-k_r x_8 - x_7 x_9 (I_x - I_z)]/I_y \\ [-k_r x_9 - x_7 x_8 (I_y - I_x)]/I_z \end{bmatrix} \tag{8}$$

The outer loop control system is selected as

$$\begin{cases} \dot{\mathbf{X}}_3 = \mathbf{A}_3 \mathbf{X}_4 \\ \dot{\mathbf{X}}_4 = \mathbf{A}_4 + \mathbf{B}_2 \mathrm{s}(\mathbf{u}_o) \end{cases} \tag{9}$$

where $\mathbf{X}_3 = [x_4, x_5]^{\mathrm{T}}, \mathbf{A}_3 = \mathrm{diag}(1,1)$, $\mathbf{X}_4 = [x_1, x_2]^{\mathrm{T}}$, $\mathbf{A}_4 = [-\frac{k_t x_1}{m}, -\frac{k_t x_2}{m}]^{\mathrm{T}}$, $\mathbf{u}_o = [u_x, u_y]^{\mathrm{T}} = [\cos x_{10} \sin x_{11}, \sin x_{10}]^{\mathrm{T}}$,where u_x and u_y can be regarded as the equivalent input signals of the horizontal control system. The specific form of the system matrix $\mathbf{B}_2$ is as follows:

$$\mathbf{B}_2 = \begin{bmatrix} \frac{\mathrm{s}(u_1)}{m} \cos(x_{12}) & \frac{\mathrm{s}(u_1)}{m} \sin(x_{12}) \\ \frac{\mathrm{s}(u_1)}{m} \cos(x_{12}) & -\frac{\mathrm{s}(u_1)}{m} \cos(x_{12}) \end{bmatrix} \tag{10}$$

By this way, (1) is decomposed into two strict-feedback fully-actuated systems. In the following context, the controllers of (6) and (9) will be designed respectively.

A. Inner loop control system

To ensure that the disturbance vector $\mathbf{D}_1$ is compensated within finite-time, the extended state observer is used to observe the disturbance signals.

In the extended state observer, define $\Omega_1 = \mathbf{X}_2$, $\Omega_2 = \mathbf{D}_1$, $\dot{\Omega}_2 = F_1(t)$. $F_1(t)$ is a vector composed of unknown but bounded functions. Based on this, the equations of the extended system can be written as

$$\begin{aligned} \dot{\Omega}_1 &= \mathbf{A}_2 + \mathbf{B}_1 \mathrm{s}(\mathbf{u}_i) + \Omega_2 \\ \dot{\Omega}_2 &= F_1(t) \end{aligned} \tag{11}$$

The estimated values of Ω_1 and Ω_2 can be represented by $\hat{\Omega}_1$ and $\hat{\Omega}_2$. The errors between the observed values and the actual values are $\tilde{\Omega}_1 = \hat{\Omega}_1 - \Omega_1, \tilde{\Omega}_2 = \hat{\Omega}_2 - \Omega_2$. Based on (11), the finite-time extended state observer is designed as

$$\dot{\hat{\Omega}}_1 = \mathbf{A}_2 + \mathbf{B}_1 \mathrm{s}(\mathbf{u}_i) + \hat{\Omega}_2 - \alpha_1 (\lceil \tilde{\Omega}_1 \rfloor^{\beta_1} + \tilde{\Omega}_1) \tag{12}$$

$$\dot{\hat{\Omega}}_2 = -\alpha_2 (\lceil \tilde{\Omega}_1 \rfloor^{\beta_2} + 2 \lceil \tilde{\Omega}_1 \rfloor^{\beta_1} + \tilde{\Omega}_1) - F_1(t) \tag{13}$$

where $\alpha_1, \alpha_2 > 0, \beta_1 \in (\frac{1}{2}, 1), \beta_2 = 2\beta_1 - 1$. $\lceil \cdot \rfloor^a$ denotes raising each element in the matrix to the a-th power. The errors dynamic of the extended state observer are

$$\dot{\tilde{\Omega}}_1 = \tilde{\Omega}_2 - \alpha_1 (\lceil \tilde{\Omega}_1 \rfloor^{\beta_1} + \tilde{\Omega}_1)$$

$$\dot{\tilde{\Omega}}_2 = -\alpha_1 (\lceil \tilde{\Omega}_1 \rfloor^{\beta_2} + 2 \lceil \tilde{\Omega}_1 \rfloor^{\beta_1} + \tilde{\Omega}_1) - F_1(t) \tag{14}$$

Proof: The error vector is chosen as follows

$$e = [\tilde{\Omega}_1^T, \tilde{\Omega}_2^T]^T \tag{15}$$

$$E = [(\lceil \tilde{\Omega}_1 \rfloor^{\beta_1} + \tilde{\Omega}_1)^T, \tilde{\Omega}_2^T]^T \tag{16}$$

The derivative of (16) is

$$e = [\tilde{\Omega}_1^T, \tilde{\Omega}_2^T]^T \tag{17}$$

$$\dot{E} = diag(|\tilde{\Omega}_1|^{\beta_1 - 1}, |\tilde{\Omega}_1|^{\beta_1 - 1}) H_1 E + H_2 E + M \tag{18}$$

where the Hurwitz matrixs $H_1 = -[\beta_1 \alpha_1 I_3, \beta_1 I_3; -\alpha_2, O_3]$, $H_2 = [-\alpha_1 I_3, I_3; -\alpha_2, O_3]$. $M = [O_3; F_1]$, $O_3 = diag(0,0,0)$. Therefore, there exists a matrix P that satisfies

$$PH_1 + H_1^T P \leq -Q_1 \tag{19}$$

$$PH_2 + H_2^T P \leq -Q_2 \tag{20}$$

where $Q_1^T = Q_1, Q_2^T = Q_2$.The Lyapunov function of the observer is chosen as

$$V_d(e) = E^T P E \tag{21}$$

Combining equation (18), the derivative of V can be obtained as

$$\dot{V}_d = E^T [diag(|\tilde{\Omega}_1|^{\beta_1 - 1}, |\tilde{\Omega}_1|^{\beta_1 - 1})(PH_1 + H_1^T P)] E^T + E^T (PH_2 + H_2^T P) E + M^T P E + E^T P M \tag{22}$$

Substituting (19) and (20) into (22), it can be known that

$$\dot{V}_d \leq -(|\tilde{\Omega}_1|_{\max})^{\beta_1-1}E^TQ_1E - E^TQ_2E + 2||E||||P||||M|| \quad (23)$$

$|\tilde{\Omega}_1|_{\max} \leq ||\tilde{\Omega}_1|| \leq ||E||^{\frac{1}{q}}, (q \in (0.5, 1))$, and combine the properties of matrices, one has:

$$\dot{V}_d \leq -\lambda_{\min}(Q_1)||E||^{3-\frac{1}{\beta_1}} - \lambda_{\min}(Q_2)||E||^2 + 2||E|| * ||P|| * ||M|| \quad (24)$$

Since both F_1 and $||\mathbf{D}_1||$ are bounded, the conclusion that $2||E||||P||||M|| \leq 2\sqrt{3}\bar{F}_1||E||||P|| \leq 2\sqrt{3}\bar{F}_1\lambda_{\min}P^{-\frac{1}{2}}V_d^{\frac{1}{2}}||P||$. Then

$$\dot{V}_d \leq -\gamma_1 V_d^{\frac{3}{2}-\frac{1}{2\beta_1}} - \gamma_2 V_d + \gamma_3 V_d^{\frac{1}{2}} \quad (25)$$

where $\gamma_1 = \lambda_{\min}(Q_1)\lambda_{\max}(P)^{\frac{3}{2}-\frac{1}{2\beta_1}}, \gamma_2 = \lambda_{\min}(Q_2)\lambda_{\max}(P)^{-1}$, $\gamma_3 = 2\sqrt{3}\bar{F}_1\lambda_{\min}(P)^{-\frac{1}{2}}||P||$. It can be known from the conclusion of Lemma 2 that the Extended State Observer (13) can provide a disturbance estimation vector $\hat{\Omega}_2 = [d_{h1}, d_{h2}, d_{h3}, d_{h4}]^T$ for the system, and the estimation errors converge to the neighborhood around the equilibrium point within finite-time. The convergence time T_d and convergence domain S_d satisfy:

$$T_d \leq \frac{\ln(1 + (\gamma_2 - \epsilon_2)V_d(e)^{\frac{1-\beta_1}{2\beta_1}}/(\gamma_1 - \epsilon_1))}{(\gamma_2 - \epsilon_2)^{\frac{1-\beta_1}{2\beta_1}}} \quad (26)$$

$$S_d = \{e_0 | \gamma_1 V_d(e)^{\frac{2\beta_1-1}{2\beta_1}} + \gamma_2 V_d(e)^{\frac{1}{2}} < \gamma_3\} \quad (27)$$

where $\epsilon_1 \in (0, \gamma_1)$, $\epsilon_2 \in (0, \gamma_1)$ are positive constant.

Based on the above proof, the observer (13) can compensate for disturbances within finite-time. Next, the design of the finite-time controller is carried out based on backstepping control. The following coordinate transformation is necessary:

$$\mathbf{z}_1 = \mathbf{X}_1 - \mathbf{X}_{1d} \quad (28)$$

$$\mathbf{z}_2 = \mathbf{X}_2 - \bar{\varsigma}_1 \quad (29)$$

where $\mathbf{X}_{1d}$ denotes the expected signals of the inner loop. To reduce the computational burden of the system, a first-order low-pass filter is incorporated into the controller. $\bar{\varsigma}_1$ is the output of the filter, and the filter is

$$\bar{\varsigma}_1 = \varsigma_1 - \sigma_1\dot{\bar{\varsigma}}_1 \quad (30)$$

where $\sigma_1 = diag(\sigma_{11}, \sigma_{12}, \sigma_{13}, \sigma_{14})$, whose diagonal elements are design parameters. ς_1 is the virtual control law in backstepping control. The application of the filter will introduce filter errors($\bar{\varsigma}_1 - \varsigma_1$). A compensation mechanism is used to ensure that the impact of filter errors on the system is eliminated within finite-time.

$$\dot{\xi}_1 = -m_1\xi_1 + \mathbf{A}_1\bar{\varsigma}_1 - \mathbf{A}_1\varsigma_1 + \mathbf{A}_1\xi_2 - \mathbf{P}_1 l_1 \quad (31)$$

$$\dot{\xi}_2 = -m_2\xi_2 - \mathbf{A}_1\xi_1 - \mathbf{Q}_1 l_2 \quad (32)$$

where ξ_1, ξ_2 are compensation signals, $\mathbf{P}_1, \mathbf{Q}_1$ are diagonal matrices, whose diagonal elements are the signs of the compensation signals. m_1, m_2, l_1, l_2 are the positive design parameters.

Due to the existence of the compensation mechanism, it is necessary to define the error transformations

$$\chi_1 = \mathbf{z}_1 - \xi_1 \quad (33)$$

$$\chi_2 = \mathbf{z}_2 - \xi_2 \quad (34)$$

Step 1:

Take the derivative of the error vector χ_1, one gets

$$\dot{\chi}_1 = \mathbf{A}_1\chi_2 - \dot{\mathbf{X}}_{1d} + \mathbf{A}_1\varsigma_1 + m_1\xi_1 + \mathbf{P}_1 l_1 \quad (35)$$

A Lyapunov function is chosen

$$V_1 = \frac{1}{2}\chi_1^T\chi_1 \quad (36)$$

Taking the derivative yields

$$\dot{V}_1 = \chi_1^T(\mathbf{A}_1\chi_2 - \dot{\mathbf{X}}_{1d} + \mathbf{A}_1\varsigma_1 + m_1\xi_1 + \mathbf{P}_1 l_1) \quad (37)$$

According to (37), the virtual control law is designed as

$$\varsigma_1 = \mathbf{A}_1^{-1}(-m_1\mathbf{z}_1 + \dot{\mathbf{X}}_{1d} - c_1\lceil\chi_1\rfloor^{2p-1}) \quad (38)$$

where $c_1 > 0$, $p > 0.5$. Substituting (38) into (37), one can get

$$\dot{V}_1 = -m_1\chi_1^T\chi_1 + \chi_1^T\mathbf{A}_1\chi_2 - c_1(\chi_1^T\chi_1)^p + \chi_1^T\mathbf{P}_1 l_1 \quad (39)$$

Step 2:

The time derivative of χ_2 can be obtained as:

$$\dot{\chi}_2 = \mathbf{A}_2 + \mathbf{B}_1\mathrm{s}(\mathbf{u}_i) + \mathbf{D}_1 - \dot{\bar{\varsigma}}_1 + m_2\xi_2 + \mathbf{A}_1\xi_1 + \mathbf{Q}_1 l_2 \quad (40)$$

A Lyapunov function for the entire inner loop can be chosen as:

$$V_i = V_1 + \frac{1}{2}\chi_2^T\chi_2 \quad (41)$$

According to (40) and (41), one has

$$\dot{V}_i = \dot{V}_1 + \chi_2^T[\mathbf{A}_2 + \mathbf{B}_1\mathrm{s}(\mathbf{u}_i) + \mathbf{D}_1 - \dot{\bar{\varsigma}}_1 + m_2\xi_2 + \mathbf{A}_1\xi_1 + \mathbf{Q}_1 l_2] \quad (42)$$

In (42), the error vector between the actual input $\mathbf{u}_i$ and the saturated input $\mathrm{s}(\mathbf{u}_i)$ is defined as

$$\mathbf{T}_1 = \mathrm{s}(\mathbf{u}_i) - \mathbf{u}_i \quad (43)$$

The actual control law appears in (43), and is designed as

$$\mathbf{u}_i = \mathbf{B}_1^{-1}(-\mathbf{A}_2 - \hat{\Omega}_2 + \dot{\bar{\varsigma}}_1 - m_2\mathbf{z}_2 - \frac{3}{2}\chi_2 - \frac{1}{2}\chi_1 - \mathbf{A}_1\mathbf{z}_1 - c_2\lceil\chi_2\rfloor^{2p-1}) \quad (44)$$

where $c_2 > 0$.

B. Outer controller design

The structures of the outer-loop controller and the inner-loop controller are similar. This section directly presents the outer-loop controller.

Introduce the coordinate transformation as

$$\mathbf{z}_3 = \mathbf{X}_3 - \mathbf{X}_{3d} \quad (45)$$

$$\mathbf{z}_4 = \mathbf{X}_4 - \bar{\varsigma}_2 \quad (46)$$

$$\chi_3 = \mathbf{z}_3 - \xi_3 \quad (47)$$

$$\chi_4 = \mathbf{z}_4 - \xi_4 \quad (48)$$

$\mathbf{X}_{3d}$ is the desired signal vector of the outer loop. $\bar{\varsigma}_2$ is a filter output. Define the Lyapunov function of the outer

loop as $V_o = \frac{1}{2}\chi_3^T\chi_3 + \frac{1}{2}\chi_4^T\chi_4$. The command filter and the compensation mechanisms with finite-time convergence are designed as:

$$\begin{aligned}\bar{\varsigma}_2 &= \varsigma_2 - \sigma_2\dot{\bar{\varsigma}}_2 && (49)\\ \dot{\xi}_3 &= -m_3\xi_3 + \mathbf{A}_3\bar{\varsigma}_2 && (50)\\ &\quad -\mathbf{A}_3\varsigma_2 + \mathbf{A}_3\xi_4 - \mathbf{P}_2 l_3 \\ \dot{\xi}_4 &= -m_4\xi_4 - \mathbf{A}_3\xi_3 - \mathbf{Q}_2 l_4 && (51)\end{aligned}$$

where $m_3, m_4, l_3, l_4 > 0, \mathbf{P}_2$ and $\mathbf{Q}_2$ are the sign function matrices of ξ_3 and ξ_4, respectively.

The virtual control law and actual control law of the outer-loop control system are

$$\varsigma_2 = \mathbf{A}_3^{-1}(-m_3\mathbf{z}_3 + \dot{\mathbf{X}}_{3d} - c_3\lceil\chi_3\rfloor^{2p-1}) \quad (52)$$

$$\begin{aligned}\mathbf{u}_o &= \mathbf{B}_2^{-1}(-\mathbf{A}_4 + \dot{\bar{\varsigma}}_2 - m_4\mathbf{z}_4 - \frac{3}{2}\chi_4 \\ &\quad -\frac{1}{2}\chi_3 - \mathbf{A}_3\mathbf{z}_3 - c_4\lceil\chi_4\rfloor^{2p-1}) && (53)\end{aligned}$$

where $c_3, c_4 > 0$. Define $\mathbf{T}_2 = \mathrm{s}(\mathbf{u}_o) - \mathbf{u}_o$.

C. Analysis of stability

Theorem 1 For system (1) with Assumpions 1-3, under the action of the extended state observer(13), the filter and its compensation mechanism (30), (31), (32), (49), (50), (51), the virtual control law (38), (52), and the actual control law (44), (53), all states of the system can converge to a small neighborhood around the equilibrium point within finite-time.

Proof A total Lyapunov function is constructed as

$$V_s = V_i + V_o \quad (54)$$

The derivation of V_s can be obtained as

$$\begin{aligned}\dot{V}_s &= -\sum_{i=1}^{4} m_i\chi_i - \sum_{i=1}^{4} c_i\chi_i^{2p} + \chi_1^{\mathrm{T}}\mathbf{P}l_1 \\ &\quad +\chi_3^{\mathrm{T}}\mathbf{P}_2 l_3 + \chi_2^{\mathrm{T}}\mathbf{Q}_1 l_2 + \chi_4^{\mathrm{T}}\mathbf{Q}_2 l_4 && (55)\\ &\quad +\chi_2^{\mathrm{T}}\tilde{\Omega}_2 + \chi_2^{\mathrm{T}}\mathbf{B}_1\mathbf{T}_1 + \chi_3^{\mathrm{T}}\mathbf{B}_2\mathbf{T}_2 \\ &\quad -\frac{3}{2}\chi_2^T\chi_2 - \frac{1}{2}\chi_1^T\chi_1 \\ &\quad -\frac{3}{2}\chi_4^T\chi_4 - \frac{1}{2}\chi_3^T\chi_3\end{aligned}$$

According to young's inequality, one has

$$\begin{aligned}\chi_1^{\mathrm{T}}\mathbf{P}l_1 &\le \frac{1}{2}\chi_1^{\mathrm{T}}\chi_1 + \frac{1}{2}l_1^2 \\ \chi_2^{\mathrm{T}}\mathbf{Q}l_2 &\le \frac{1}{2}\chi_2^{\mathrm{T}}\chi_2 + \frac{1}{2}l_2^2 \\ \chi_2^{\mathrm{T}}\tilde{\Omega}_2 &\le \frac{1}{2}\chi_2^{\mathrm{T}}\chi_2 + \frac{1}{2}||\tilde{\Omega}_2||^2 \\ \chi_2^{\mathrm{T}}\mathbf{B}_1\mathbf{T}_1 &\le \frac{1}{2}\chi_2^{\mathrm{T}}\chi_2 + \frac{||\mathbf{B}_1\mathbf{T}_{1\max}||^2}{2} \\ \chi_3^{\mathrm{T}}\mathbf{P}_2 l_3 &\le \frac{1}{2}\chi_3^{\mathrm{T}}\chi_3 + \frac{1}{2}l_3^2 && (56)\\ \chi_4^{\mathrm{T}}\mathbf{Q}_2 l_4 &\le \frac{1}{2}\chi_4^{\mathrm{T}}\chi_4 + \frac{1}{2}l_4^2 \\ \chi_4^{\mathrm{T}}\mathbf{B}_2\mathbf{T}_2 &\le \frac{1}{2}\chi_4^{\mathrm{T}}\chi_4 + \frac{||\mathbf{B}_2\mathbf{T}_{2\max}||^2}{2}\end{aligned}$$

Substituting (56) into (55) arrives at

$$\begin{aligned}\dot{V}_s &\le -\sum_{i=1}^{4} m_i(\chi_i^{\mathrm{T}}\chi_i) - \sum_{i=1}^{4} c_i(\chi_i^{\mathrm{T}}\chi_i)^p \\ &\quad +\sum_{i=1}^{4}(\frac{1}{2}l_i^2) + \frac{1}{2}\tilde{\mathbf{D}}_1^{\mathrm{T}}\tilde{\mathbf{D}}_1 \\ &\quad +\frac{||\mathbf{B}_1\mathbf{T}_{1\max}||^2 + ||\mathbf{B}_2\mathbf{T}_{2\max}||^2}{2} && (57)\end{aligned}$$

According Lemma 1, the following inequality is true

$$\dot{V}_s \le -A_{s1}V_s - A_{s2}V_s^p + B_s \quad (58)$$

where

$$\begin{aligned}A_{s1} &= 2\min\{m_i(i=1,2,3,4)\}, && (59)\\ A_{s2} &= 2^p\min\{c_i(i=1,2,3,4)\} && (60)\\ B_s &= \frac{1}{2}\tilde{\Omega}_2^2 + \\ &\quad \frac{||\mathbf{B}_1\mathbf{T}_{1\max}||^2 + ||\mathbf{B}_2\mathbf{T}_{2\max}||^2}{2} && (61)\end{aligned}$$

(58) can be equivalently transformed into

$$\dot{V}_s \le -\kappa A_{s1}V_s - (1-\kappa)A_{s1}V_s - A_{s2}V_s^p + B_s \quad (62)$$

or

$$\dot{V}_s \le -A_{s1}V_s - \kappa A_{s2}V_s^p - (1-\kappa)V_s^p + B_s \quad (63)$$

where $0 < \kappa < 1$. If (62) is true, the conclusion of Lemma 2 indicates that the tracking error will converge to $V_s \le \frac{B_s}{(1-\kappa)A_{s1}}$, and the convergence time is

$$T_{s1} \le \frac{1}{\kappa A_{s1}p}\ln\frac{[\kappa A_{s1}V_s^p(0) + A_{s2}]}{A_{s2}} \quad (64)$$

where $V_s(0)$ represents the energy of the initial state.

If inequality (63) holds, review Lemma 2, one can directly obtain that the error will converge to $V_s^p \le \frac{B_s}{(1-\kappa)A_{s2}}$ in a finite time. the convergence time is

$$T_{s2} \le \frac{1}{A_{s1}p}\ln[A_{s1}V_s^p(0) + \kappa A_{s2}]\frac{[A_{s1}V_s^p(0) + \kappa A_{s2}]}{\kappa A_{s2}} \quad (65)$$

Based on the above analysis, the errors satisfie $\chi = \{(\chi_1, \chi_2, \chi_3, \chi_4)|V_s^p \le \max\{\frac{B_s}{(1-\kappa)A_{s1}}, \frac{B_s}{(1-\kappa)A_{s2}}\}\}$. The errors χ converge within a finite-time $T_s = \max\{T_{s1}, T_{s2}\}$.

According to the definition of the tracking errors: $\chi_i = \mathbf{z}_i - \xi_i$, if the conclusion that ξ_i converges is true, $\mathbf{z}_i$ also converges. The proof with the convergence of ξ_i is given as follows:

A Lyapunov function about ξ_i is selected as

$$V_\xi = \sum_{i=1}^{4}\frac{1}{2}\xi_i^T\xi_i \quad (66)$$

Taking the derivative of V_ξ, one gets

$$\begin{aligned}\dot{V}_\xi &= \sum_{i=1}^{4} - m_i\xi_i^T\xi_i + \xi_1^{\mathrm{T}}\mathbf{A}_1(\bar{\varsigma}_1 - \varsigma_1) + \xi_3^{\mathrm{T}}\mathbf{A}_3(\bar{\varsigma}_2 - \varsigma_2) \\ &\quad -\sum_{i=1}^{4}(l_{1i}|\xi_{1i}| - l_{2i}|\xi_{2i}|) \\ &\quad -\sum_{j=1}^{2}(l_{3i}|\xi_{3i}| - l_{4i}|\xi_{4i}|)\end{aligned} \tag{67}$$

Inspired by [22], if an appropriate l is selected, the following inequality can be made to hold.

$$\xi_1^{\mathrm{T}}\mathbf{A}_1(\bar{\varsigma}_1 - \varsigma_1) \leq \sum_{i=1}^{4}\bar{\varpi}_i \tag{68}$$

$$\xi_3^{\mathrm{T}}\mathbf{A}_3(\bar{\varsigma}_2 - \varsigma_2) \leq \sum_{j=1}^{2}\breve{\varpi}_j \tag{69}$$

where $\bar{\varpi}_i, \breve{\varpi}_j$ are bounded.

Substitute (68) and (69) into (67), we have

$$\begin{aligned}\dot{V}_\xi &\leq \sum_{i=1}^{4}(-m_i\xi_i^T\xi_i) - l_2\xi_2 - l_4\xi_4 \\ &\quad -\sum_{i=1}^{4}(l_{1i}|\xi_{1i}| - \bar{\varpi}_i) - \sum_{i=1}^{2}(l_{3i}|\xi_{3i}| - \breve{\varpi}_i) \\ &\leq -A_{\xi 1}V_\xi - A_{\xi 2}V_\xi^{\frac{1}{2}}\end{aligned} \tag{70}$$

where $A_{\xi 1} = 2\min\{m_i(i = 1,2,3,4)\}$, $A_{\xi 2} = \sqrt{2}\min\{l_{1i} - \bar{\varpi}_i, l_{3i} - \breve{\varpi}_i\}$ are positive. Review Lemma 2, ξ_i can also converge to equilibrium in a finite-time T_ξ. Up to this point, the conclusion that $\mathbf{z}_i$ can converge to the neighborhood around the equilibrium point within finite-time $T_s \leq \max\{T_d, T_s, T_\xi\}$ can be known. The proof is completed.

IV. Numerical Simulation Experiment

The numerical simulation in this paper is completed using MATLAB software. It is assumed that the structural parameters of a quadrotor UAV are as follows: $m = 2.4\mathrm{kg}$, $g = 10\mathrm{N/s}^2$, $I_x = I_y = 0.2\mathrm{kg}\cdot\mathrm{m}^2$, $I_z = 0.3\mathrm{kg}\cdot\mathrm{m}^2$, $k_t = k_r = 0.006\mathrm{N}\cdot\mathrm{m/s}$. The selection of all control parameters can refer to Table 1.

Remark 2 The principles for parameter selection are as follows: The maximum total lift and the maximum input torque are given based on the structural parameters of the UAV. Setting larger observer parameters α_1, α_2 can enhance the speed at which the observer approximates disturbance signals, thereby improving the robustness of the system. Under ideal circumstances, the filter parameters σ_{1i}, σ_{2i} should be close to the sampling period. Providing a smaller control gains c_1, c_2, c_3,and c_4 help the attitude angle track the inverse solution signals provided by the outer loop more easily. The selection of the remaining parameters has little significant impact on the control effect, so there is greater flexibility in selection.

The quadrotor UAV is required to fly along a cardioid in the air. Given Reference Signal: $\mathbf{X}_{3d} = [2\cos(0.2t) - \cos(0.4t),\ 2\sin(0.2t) - \sin(0.4t)]^T$. The flight altitude is 5 meters. Set the simulation time to 60s. When t = 10s, the external disturbance signals are applied in the system, and its specific expression is: $\mathbf{D}_1 = [0,\ 0.4\cos(t) + 0.2\sin(0.5t),\ 0.4\sin(t),\ 0.2\cos(2t) + 0.1\sin(t)]^T$. The initial values of all variables are set to 0. [23] uses a neural network disturbance observer to compensate for disturbance signals. This scheme is selected for comparison with the scheme proposed in this paper. The program runtime is completed within three seconds. All of the numerical simulation results are shown in Fig.2-Fig.8.

TABLE I
Control Parameters

Parameter	Value
α_1	5
α_2	10
β_1	0.8
$\sigma_{1i}, \sigma_{2i} (i = 1,2,3,4)$	0.02
m_1	2.2
m_2	1.4
m_3	7.2
m_4	2.3
$l_i (i = 1,2,3,4)$	0.1
c_1	1.7
c_2	0.08
c_3	0.26
c_4	0.06
$u_{1\max}$	32
$u_{2\max}$	5
$u_{3\max}$	5
$u_{4\max}$	5
p	0.9

Fig.2 shows the output curves of the inner-loop system. All signals can track the given reference signals within 4.5 seconds. The inner-loop control system can accurately track the desired signal provided by the outer loop, which means the UAV can complete the trajectory tracking task. After the fault occurs, the system exhibits slight chattering and does not deviate from the desired trajectory. The control effect is good.

Fig.3 is the output curves of the outer-loop system. It can be seen that the system can accurately track the given reference signal within 2.5 seconds. In addition, in the presence of external disturbances, the UAV does not deviate from the desired trajectory.

Fig.4 shows the flight trajectory of the UAV in three-dimensional space. Among them, the blue line represents the trajectory of the scheme proposed in this paper, and the red line represents the flight trajectory of the scheme in [23]. It can be seen from the figure that the overshoot of the proposed scheme is smaller. Notably, in the presence of disturbances, the scheme in [23] exhibits obvious chattering; in contrast, the chattering phenomenon of the proposed scheme is not obvious.

The tracking effect of the disturbance observer is shown in Fig.5. Due to the high coupling inherent in the system, the observer will generate slight fluctuations in the initial stage, and this phenomenon will not affect the control effect. If the disturbances occur, the observer can quickly compensate for disturbance signals, which means it can provide the controller with rapid and accurate estimates of disturbances.

Fig. 6 and Fig. 7 show the response curves of the inner-loop compensation signals and the outer-loop compensation signals, respectively. Within four seconds, all compensation

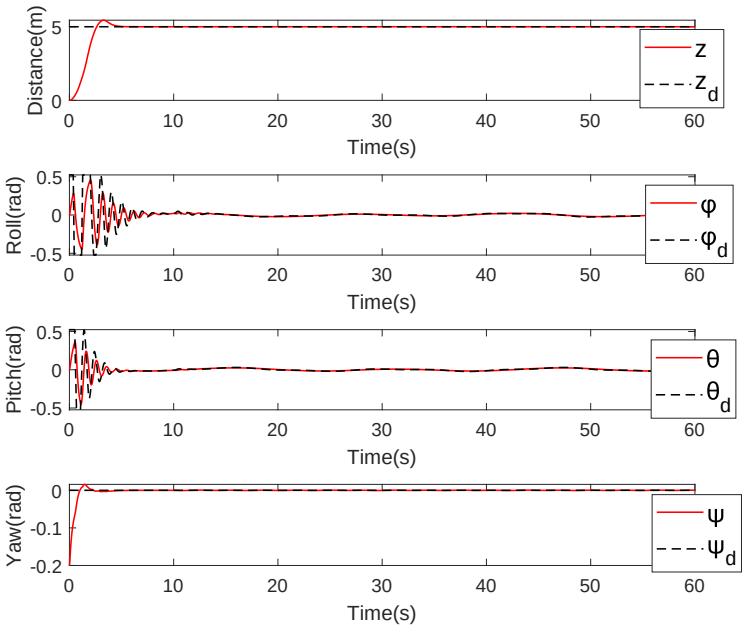

Fig. 2. Inner-loop Control Effect.

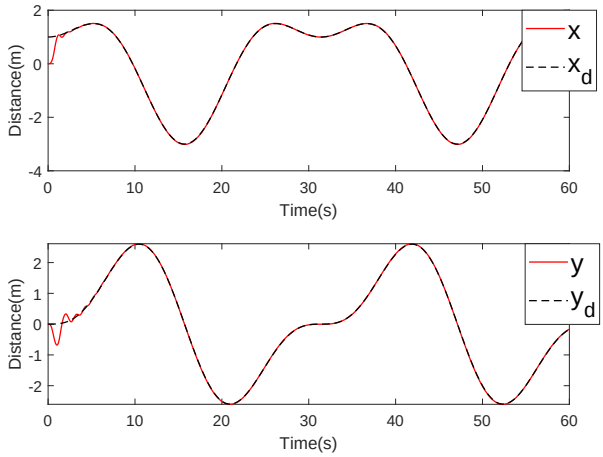

Fig. 3. Outer-loop Control Effect.

signals converge to the neighborhood near the equilibrium point, which indicates that the compensator has reached a steady state.

Finally, Fig.8 shows the input signals of the quadrotor UAV. Due to the strong coupling between the roll angle and the pitch angle, the two angles are inevitably affected by each other in the process of tracking their respective reference signals. As a result, chattering occurs during the transient process, which is acceptable.

V. Conclusion

A novel finite-time trajectory tracking control scheme for the quadrotor UAVs is proposed in this paper. Based on Lyapunov theory, both the proposed controller and observer are proven to achieve finite-time convergence. The introduction of command filtering technology not only avoids the differentiation of complex matrices, but also achieves the approximation of the derivative of the virtual control law within a finite time. Numerical simulation results demonstrate that the proposed scheme exhibits excellent control performance, with no variables showing divergence. For future work, we will adopt optimization algorithms to seek more appropriate control parameters.

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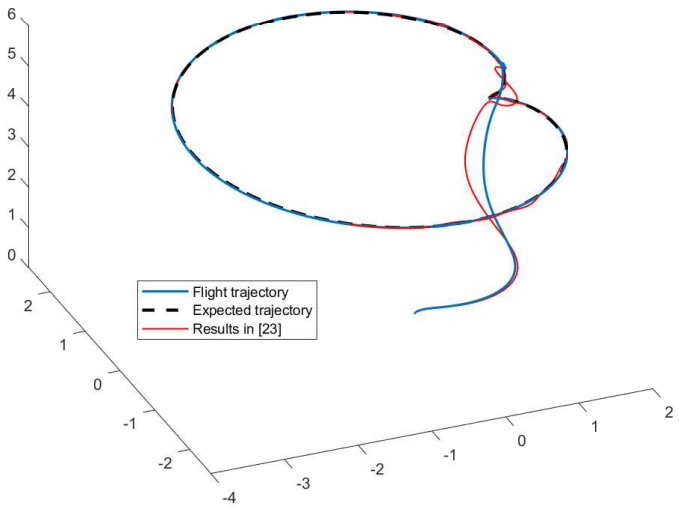

Fig. 4. Flight Trajectory.

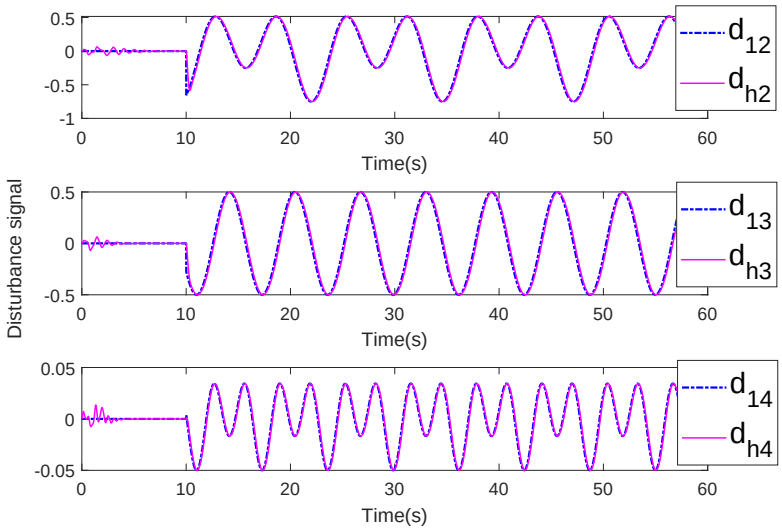

Fig. 5. Disturbance Signal Compensation Effect.

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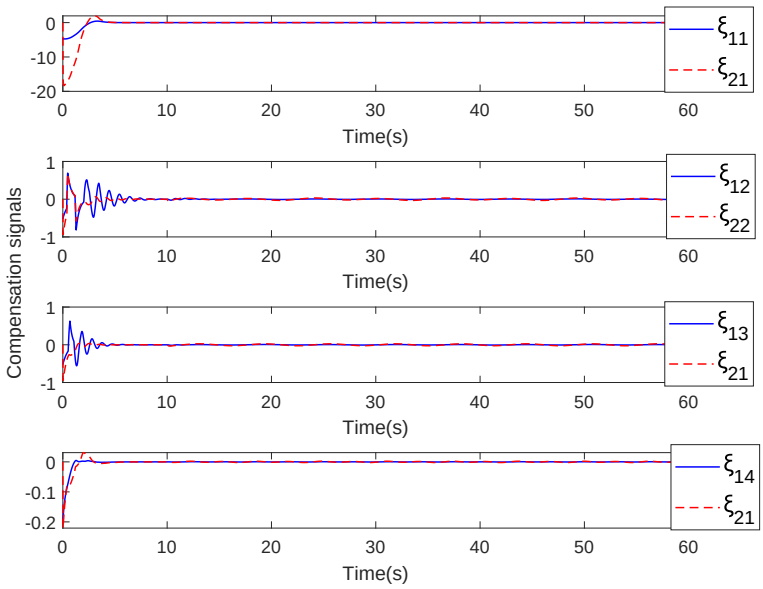

Fig. 6. Inner-loop Filtering Compensation Signals.

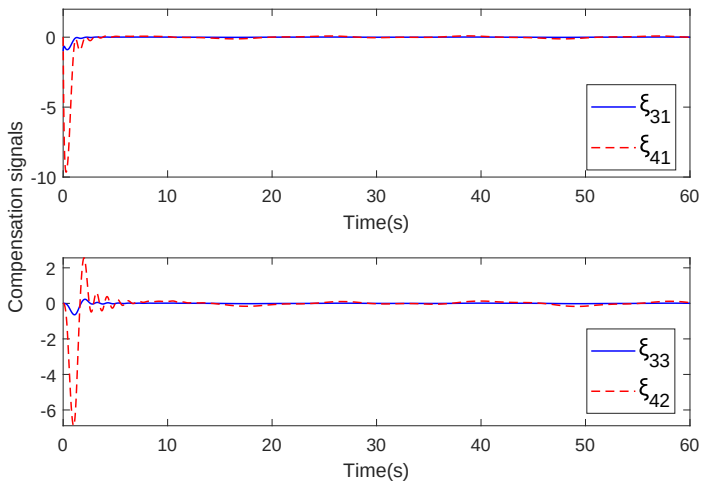

Fig. 7. Outer-loop Filtering Compensation Signals.

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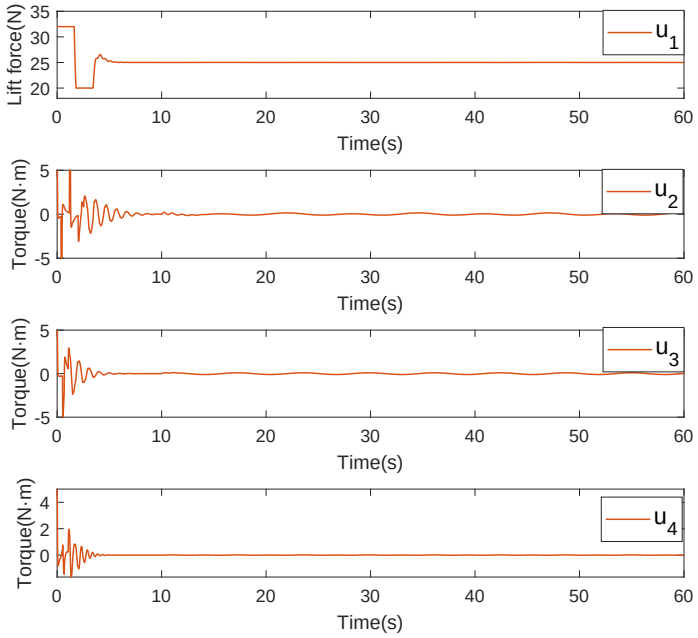

Fig. 8. Input Signals.